

Oxide-Based RRAM: Unified Microscopic Principle for both Unipolar and Bipolar Switching

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Abstract

A unified microscopic principle is proposed to clarify resistive switching behaviors of transition metal oxide based resistive random access memories (RRAM) for the first time. In this unified microscopic principle, both unipolar and bipolar switching characteristics of RRAM are correlated with the distribution of localized oxygen vacancies in the oxide switching layer, which is governed by the generation and recombination with dissociative oxygen ions. Based on the proposed microscopic principle, an atomistic simulation method is developed to evaluate critical memory performance, and successfully conduct the device optimization. The experimental data are well in line with the developed simulation method.

Introduction

Metal oxide based resistive switching memory or resistive random access memory (RRAM) has demonstrated superior performance and scalability [1-3]. Various switching phenomena, such as unipolar and bipolar switching behaviors, are usually explained by different physical mechanisms [4-8]. However, recent reports showed that both unipolar and bipolar switching can co-exist in the same devices [3,9-11], indicating these two behaviors would share the similar physical origin [10-12]. In this paper, a unified microscopic principle is proposed to clarify both unipolar and bipolar switching behaviors in metal oxide based RRAM for the first time. In this unified microscopic principle, switching characteristics are correlated with oxygen vacancy (Vo) distribution in the oxide switching layer, which is governed by the generation and recombination of Vo with dissociative oxygen ions (O^{2-}). By introducing transport equations of the O^{2-} and distribution of Vo, a new simulation methodology is established to quantify the switching characteristics in term of critical memory performance evaluation, and device optimization can thus be successfully conducted.

Unified Microscopic Model and Simulation Method

The unified microscopic model for both unipolar and

bipolar switching is illustrated in Fig.1. Major claims include: 1) The conduction is due to electron hopping transport through Vo and the various resistance states are correlated with the distribution of Vo [8,11]; 2) Forming/SET and RESET processes are correlated with the respective generation of Vo and recombination of positive charged Vo^{2+} (due to electron-depletion in Vo) with O^{2-} ; 3) For bipolar RESET process, the recombination is dominated by electric field controlled electron depletion in oxygen vacancy (producing Vo^{2+}) and O^{2-} migration from the cathode interface; 4) For unipolar RESET process, the recombination is dominated by electric field controlled electron depletion in oxygen vacancy (producing Vo^{2+}) and Joule heating controlled thermal releasing of O^{2-} (from the easily-reduced oxide clusters which are closely neighbored with the Vo^{2+}). Based on this microscopic principle, an atomistic simulation method is established to quantify the switching characteristics by introducing correlated equations after considering atomistic effects as in Fig.2. Table I lists the simulation parameters, where L , κ , ϵ_r , d , r , T_{ox} and S refer to the distance between nearest lattice oxygen, dielectric constant, formation energy of Vo, length of the rupture region, radius of the filament tips, oxide thickness, and device size, respectively.

Model Verification and Evaluation

To verify the proposed mechanism and models, HfO_x -based RRAM devices with different electrodes were fabricated. Detail fabrication processes were described elsewhere [13-15].

i) Conduction Mechanism: Both hopping and metallic conduction behaviors are observed in the same devices under different SET current compliances, as shown in Fig.3a&b. These measured data indicate that hopping/metallic conduction is determined by the electron transport distance between Vo, as illustrated in Fig.3c. In the meanwhile, the hopping conduction is believed to be a key factor to reduce the switching power [11]. Therefore, suppressing Vo clusters formation is desired to achieve low power.

ii) SET process: Fig.4 compares the measured distributions of SET voltage with our simulation data. The excellent agreement validates our proposed simulation method.

Comparing the simulated electric field distribution and the distributions of Vo generation probability during SET process (in respective Figs.5&6), Vo generation probability is believed to be closely related to the local electric field. Fig.7 illustrates simulated Vo concentration as a function of the depth from electrode to filament tips during SET process. As shown in Fig.8, multi-filaments or single cone-shaped filament is relatively easy to be formed at lower or higher SET voltages due to the local electric field effect. This conclusion is also supported by the measured I-V curves shown in Figs.9&10. The sharp/gradual resistance reductions at respective higher/lower SET voltages are associated with the formation of single thick-filament and multi thin-filaments, respectively. In Fig.10a, sharp transition occurs when the 5th pulse is applied. On the other hand, after reducing pulse height, uniform resistive distribution can be observed under the applied voltage with various pulses. The multiple thin-filaments could result in hopping conduction in LRS as discussed in Fig.3c. Therefore, despite of the low resistance, the hopping conduction can be still observed owing to the smaller SET voltage, as shown in Fig.11, which is beneficial to the multilevel data storage control and reduction of RESET current.

iii) RESET process: Figs.12&13 respectively depicts the simulated 3D&1D Vo distributions of the stepwise bipolar RESET process. It has been reported that O²⁻ migrated from interface to bulk oxide layer is critical for bipolar RESET process [8]. As a result, the recombination of Vo would happen near the cathode side firstly and the length of rupture region would increase with the RESET voltage [8], represented by the higher HRS by larger RESET voltage as demonstrated in Fig.14. Fig.15 shows SET voltage increases as the RESET voltage in previous cycle due to the increased rupture length, consistent with the model prediction. Fig.16 shows the simulated 3D Vo distribution probability in unipolar RESET process. Since O²⁻ are generated in the bulk oxide layer due to thermal decomposition of the easily-reduced oxide clusters, the Vo distributions are thus different from the bipolar case. The migration direction of O²⁻ is usually perpendicular to filament. Thus it is challenging to control/achieve the multilevel switching by controlling RESET voltage for the unipolar device. As illustrated in Fig.17, both the electron depletion process (P1) and the O²⁻-releasing process (P2) should occur before the Vo recombination in unipolar RESET process. Fig.18 shows RESET current as a function of SET current for unipolar switching. Two different regions can be observed, which are dominated by the respective occurring sequence of P1 and P2 at different LRS. If P1 occurs before P2, unipolar RESET current is approximately equal to bipolar type switching in the same device; whereas, unipolar RESET current is larger if P2 is prior to P1, as demonstrated in Fig.19&20, supporting the proposed mechanism.

Table II summaries the predictions based on the proposed mechanism and the developed model. From these parameters, in order to achieve high performance unipolar RRAM devices,

easily-reduced oxide layer which can easily resolve O²⁻ by thermal reaction is thus demanded; On the other hand, the cathode material is more crucial for bipolar RESET behavior. Based on the property shown in Fig.18, it is predicted that the multilevel unipolar switching could only occur under large RESET current, corresponding to the domination of the electron depletion voltage of Vo. However, it might be difficult to achieve high performance multi-bit storage (e.g. good endurance is difficult to achieve) due to the high O²⁻ loss during RESET process.

Conclusion:

For the first time, a unified microscopic principle and correlated modeling and simulation methods are proposed to quantify both unipolar and bipolar switching characteristics of oxide-based RRAM, providing a design guideline to realize the high performance unipolar and bipolar RRAM devices.

Acknowledgements:

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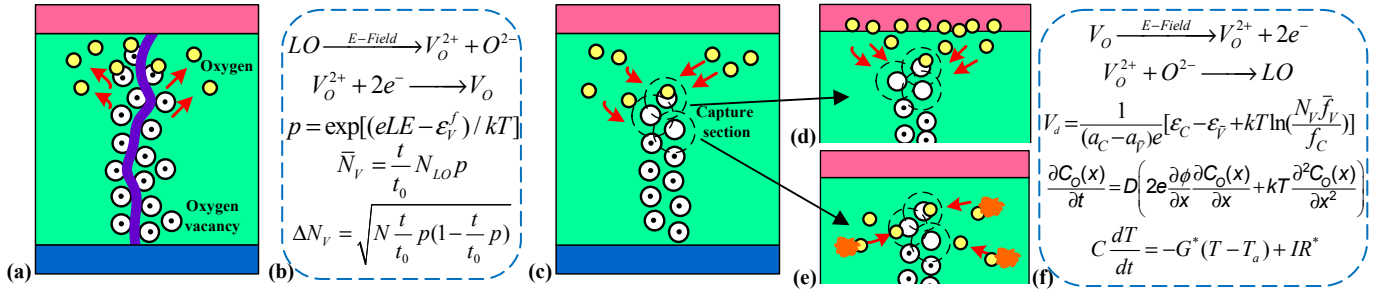


Fig.1 Schematic unified switching mechanism for both unipolar and bipolar RRAM dominated by the generation and recombination of oxygen vacancies (V_O) with dissociative oxygen ions (O^{2-}). (a) SET mechanism correlated with the generation of V_O and O^{2-} from lattice oxygens (LO) and (b) the related equations; (c) RESET mechanism correlated with the recombination of V_O due to the capture of charged oxygen vacancies (V_O^{2+}) to O^{2-} ; (d) For bipolar RESET, both electron (e^-) depletion of V_O at a critical electric field [8] induced V_O^{2+} and electric field controlled O^{2-} drift from interface O^{2-} reservoir to the capture section region dominate V_O recombination; (e) For unipolar RESET, both the electric field induced e^- depletion of V_O and the Joule heating induced thermal decomposition of O^{2-} from the easily-reduced oxide clusters or grains in the capture section region dominate V_O recombination; (f) Equations related to the e^- -depletion effect of V_O , electric field driven O^{2-} transport, and Joule heating induced thermal decomposition of O^{2-} .

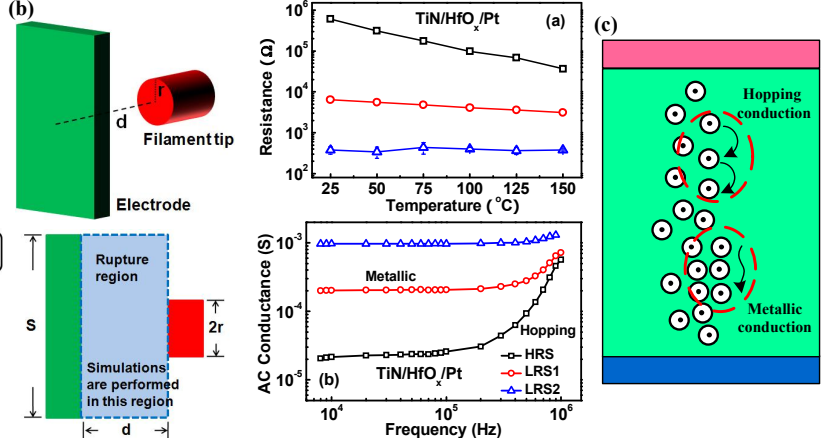
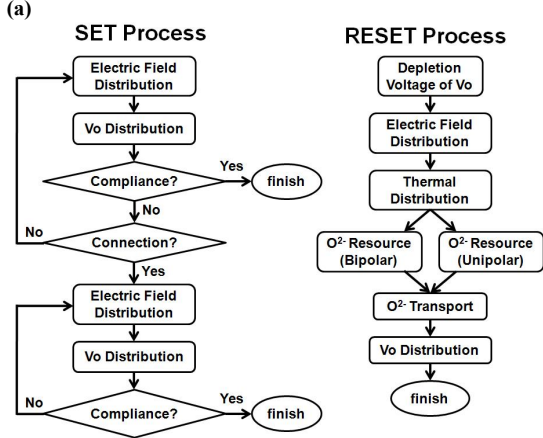


Fig.2 (a) Schematic flow chart of the 3D simulation method based on the unified switching mechanism incorporating with the equations characterized the related effects. (b) Schematic view of 3D structure of the cylinder-like filament tip for the simulation and the critical switching parameters such as the radius r of the filament, the length d of the filament gap region.

Fig.3 (a) Measured temperature dependence of resistance and (b) frequency dependence of AC conductance for the same devices at various resistive states. Both metallic-like and hopping conduction are observed. (c) Schematic conduction mechanisms, where metallic is originated from e^- transport among the cluster V_O and hopping is due to e^- transport between separated V_O .

Table I Summary of typical parameters in the simulation. The physical meanings of the symbols are explained in the text.

L	0.25 nm
κ	24
ϵ_{V_f}	0.9 eV
kT	0.026 eV
d	2 nm
r	1 nm
T_{ox}	10 nm
S	8×8 nm²

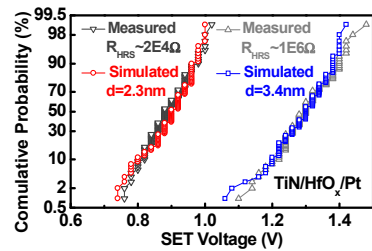


Fig.4 Simulated SET voltage distributions and the comparison with measured data. Different RESET voltages are applied to achieve different HRS resistances with different length of filament gap region.

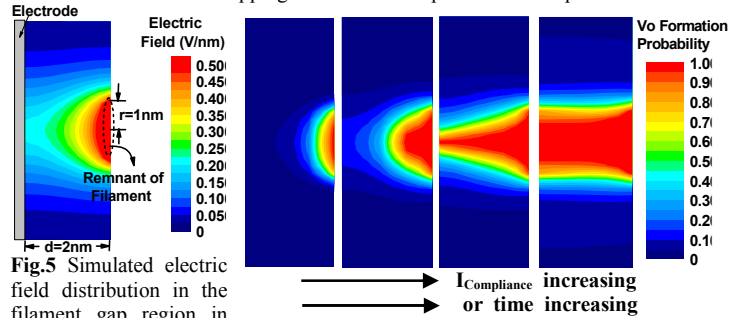


Fig.5 Simulated electric field distribution in the filament gap region in the beginning phase of SET process.

Fig.6 Simulated V_O generation probability distribution at various SET states.

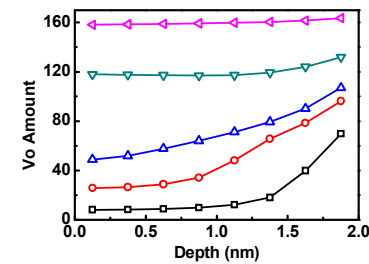


Fig.7 Simulated V_O area density as a function of depth from electrode to the front-end of the filament under different current compliance.

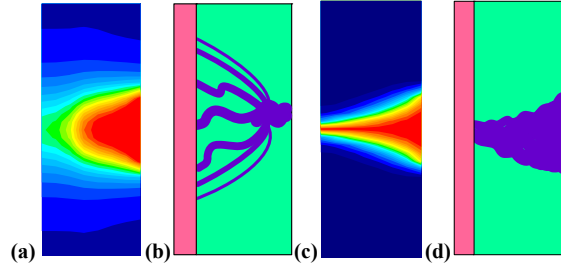


Fig.8 (a) & (c) Simulated probability distributions of generated V_O at lower and higher SET voltages. The color label is same to Fig.6. (b) & (d) Schematic geometries of the filaments formed at lower and higher SET voltages.

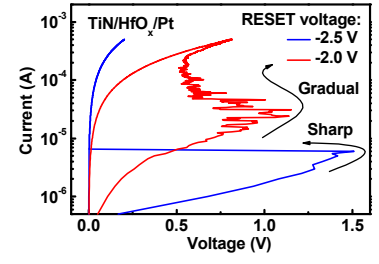


Fig. 9 Measured I-V curves during SET processes. Sharp resistance change at a higher SET voltage and gradual resistance change at a lower voltage are observed.

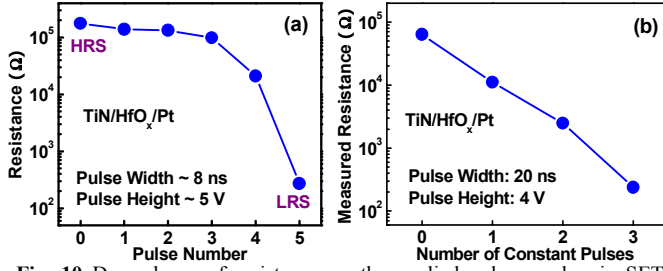


Fig. 10 Dependence of resistance on the applied pulse number in SET process. (a) Sudden resistance change occurs at the larger pulse number when narrower and larger voltage pulses are applied, and (b) nearly linearly resistance change with the pulse number is observed when the wider and smaller voltage pulses are applied.

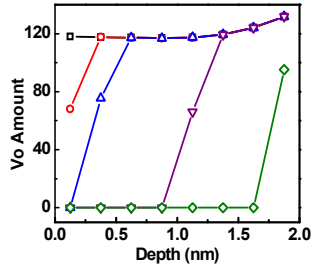


Fig. 13 Simulated Vo area density as a function of depth at different RESET voltages for bipolar switching.

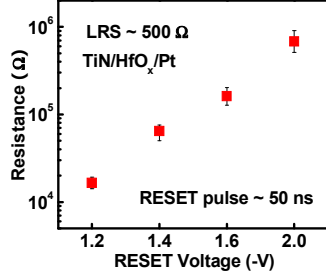


Fig. 14 Measured R_{HRS} - V_{RESET} curve. R_{HRS} increases with RESET voltage due to the increased filament gap region.

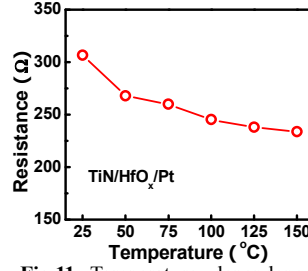


Fig. 11 Temperature dependence of LRS resistance after low voltage SET. Semiconducting-like behavior indicates hopping conduction is dominated.

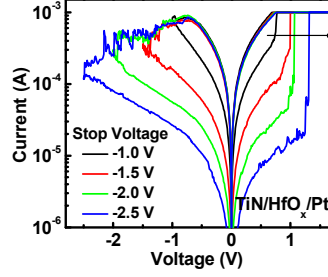


Fig. 15 Measured I-V curves using different RESET voltages. The following SET voltage increases with RESET voltage.

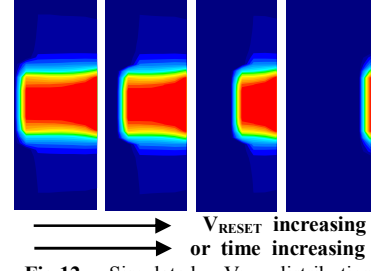


Fig. 12 Simulated Vo distribution probability at different phases during RESET process for bipolar switching. The color label is same to Fig. 6.

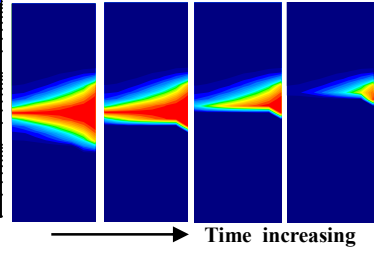


Fig. 16 Simulated Vo distribution probability at different phases during RESET process for unipolar switching. The color label is same to Fig. 6.

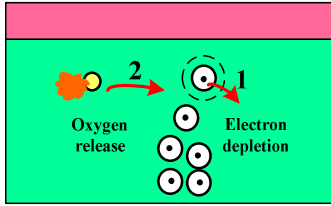


Fig. 17 Schematic of unipolar RESET. RESET process initiates only when both the e^- depletion of Vo and the thermal decomposition of O^{2-} from the easily-reduced oxide clusters or grains occur simultaneously.

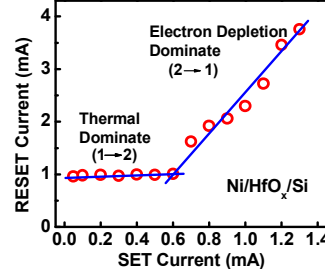


Fig. 18 Measured I_{RESET} - I_{SET} curve for unipolar switching. Two distinction regions correlated with electron depletion and thermal decomposition dominated RESET processes are observed.

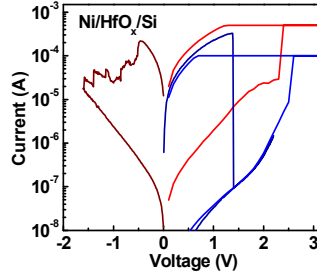


Fig. 19 Measured I-V curves of Ni/HfO_x/Si device for both unipolar and bipolar switching with nearly constant RESET currents, indicating the e^- depletion dominated RESET process for unipolar switching.

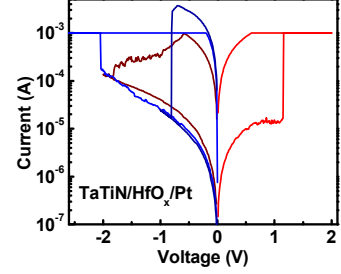


Fig. 20 Measured I-V curves of TaTiN/HfO_x/Pt devices. The higher unipolar RESET current than bipolar is observed, indicating that the thermal decomposition dominated RESET process occurs during unipolar switching.

Table II Summary of model predictions

Parameters	Related Physical Mechanisms	Impact Factors	Optimization Solution (Red: material design Blue: electrical operation)
SET voltage	Vo generation under electric-field	ϵ_v^f, d	$V_{SET} \downarrow: \epsilon_v^f \downarrow, V_{RESET} \downarrow$
SET speed	Vo generation under electric-field	$\epsilon_v^f, d, V_{SET}, RC$ delay	SET speed $\uparrow: \epsilon_v^f \downarrow, V_{SET} \uparrow$
LRS resistance	e^- transport (hopping or metallic)	Distance between Vos	Uniformity $\uparrow: \epsilon_v^f \downarrow, V_{SET} \downarrow, R_{LRS} \uparrow: Vo$ distance $\uparrow$
RESET voltage (Bi)	$Vo \rightarrow Vo^{2+}, O^{2-}$ transport/recombination	V_d, ϵ_o^{int} (interface barrier of O^{2-})	$V_{RESET} \downarrow: \epsilon_o^{int} \downarrow$
RESET voltage (Uni)	$Vo \rightarrow Vo^{2+}, O^{2-}$ recombination	V_d (depletion voltage), I_{RESET}	$V_{RESET} \downarrow: I_{RESET} \downarrow$
RESET current (Bi)	NA	V_{RESET}, R_{LRS}	$I_{RESET} \downarrow: R_{LRS} \uparrow$
RESET current (Uni)	O^{2-} thermal-release	$T_{releasing}, V_{RESET}$	$I_{RESET} \downarrow: T_{releasing} \downarrow, R_{LRS} \uparrow$
RESET speed (Bi)	O^{2-} migration/recombination	D (diffusion coefficient), V_{RESET}	RESET speed $\uparrow: D \uparrow, V_{RESET} \uparrow$
RESET speed (Uni)	O^{2-} thermal-release/recombination	G (thermal conductivity), V_{RESET}	RESET speed $\uparrow: D \uparrow, G \downarrow, V_{RESET} \uparrow$
HRS resistance	e^- transport (hopping or metallic)	d	Uniformity $\uparrow: R_{LRS}$ uniformity $\uparrow, R_{HRS} \uparrow: V_{RESET} \uparrow$
Endurance (Bi)	Consumption of interface O^{2-}	ϵ_v^f, N_o^{int} (interface O^{2-} amount)	Endurance $\uparrow: \epsilon_v^f \uparrow, N_o^{int} \uparrow$, adding interface layer
Endurance (Uni)	Consumption of easily-reduced clusters	N_o^{bulk} (bulk O^{2-} amount)	Endurance $\uparrow: N_o^{bulk} \uparrow$, adding resolvable layer
Retention	Vo gen/rec/diffusion without E-Field	$\epsilon_v^f, \epsilon_v^d$ (diffusion barrier of Vo)	Retention $\uparrow: \epsilon_v^f \uparrow, \epsilon_v^d \uparrow$
Multilevel (Bi)	Change of filament gap length / Vo amount	d	$V_{SET} \downarrow$
Multilevel (Uni)	Change of filament gap length / Vo amount	d	adding resolvable layer, $I_{RESET} \uparrow$